3D Gaussian Splatting for Industrial Machine Reconstruction

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Abstract

This report presents a complete implementation of 3D Gaussian Splatting (3DGS) for the photorealistic reconstruction of a Caterpillar 950F wheel loader a complex industrial machine exhibiting metallic surfaces, glass reflections, and fine structural detail. In line with the methodology by Kerbl et al. (2023), the pipeline begins with Structure-from-Motion (SfM) using COLMAP, where the entire set of 150 input images is aligned with a reprojection error of 0.454 px and 93,611 sparse 3D points are estimated. Subsequently, these points are employed to initialize a scene with learnable anisotropic 3D Gaussian primitives, optimized for 30,000 iterations on a cloud-based NVIDIA Tesla T4 GPU. Consequently, a model with a PSNR of 31.53 dB is obtained, which can be considered a very high-fidelity reconstruction of an outdoor industrial object. This report describes the theoretical foundations, anisotropic covariance modelling, full experimental pipeline, dataset-specific surface observations, and analysis of limitations and future improvements.

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1. Introduction

The ability to turn photos of real-world objects into 3D models is a common problem in computer vision. It has many uses, including in robotics, autonomous inspection, construction monitoring, and industrial digital twins. Traditional photogrammetric pipelines create accurate geometry, but they need significant manual work and do not support photorealistic real-time rendering by default. Neural scene representations have completely changed this field in the last few years. Neural Radiance Fields, introduced by Mildenhall et al. (2020), achieved incredible photorealism by encoding scenes as continuous volumetric functions in a multilayer perceptron and rendering them using ray-marching. NeRF, on the other hand, requires up to 48 hours of training and produces frames only every few seconds, which is not real-time. Figure 1 shows that older fast methods like InstantNGP and Plenoxels trade quality for speed.

Figure 1 (from Kerbl et al., 2023): Comparison of novel-view synthesis methods. 3DGS achieves 135 FPS at quality equal to or better than Mip-NeRF360 (48h training), while requiring only 6-51 minutes to train.

In 2023, Kerbl et al. released 3D Gaussian Splatting (3DGS), which uses GPU-accelerated tile-based rasterization to create photorealistic images in real time by showing scenes as clear groups of anisotropic 3D Gaussian primitives. This gets rid of the need for ray-marching and makes rendering speeds of over 100 FPS possible.

This project uses the 3DGS pipeline to rebuild a Caterpillar 950F wheel loader. This is a difficult industrial subject because it has shiny metal surfaces, clear glass, large areas of paint that are all the same colour, and small structural parts. The aim is to create a high-quality 3D reconstruction and carefully look at how 3DGS works on surfaces that don't follow the usual Lambertian reflectance rules.

2. Theoretical Background

2.1 Overview of the 3DGS Pipeline

The 3DGS pipeline (see Figure 2) begins with a sparse SfM point cloud and ends with a collection of 3D Gaussians. In training, a fast tile-based renderer is used to jointly optimize and adaptively control these in terms of density. The same renderer can be used for real-time novel-view synthesis after it has been trained.

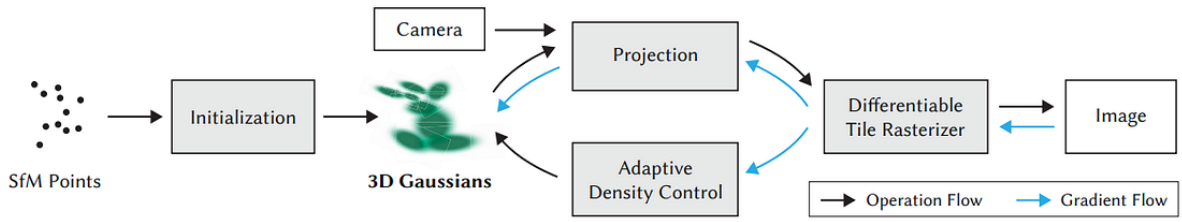

Figure 2 (from Kerbl et al.): Overview of the 3DGS optimisation pipeline. The method iterates between rendering with the tile-based rasterizer and updating Gaussian parameters via backpropagation, with periodic adaptive density control.

2.2 3D Gaussian Representation

Each 3D Gaussian primitive G_i has four groups of learnable parameters: a mean position $\mu_i \in \mathbb{R}^3$ in world space; a 3×3 covariance matrix Σ_i that shows shape, size, and orientation; an opacity $\alpha_i \in [0,1]$; and spherical harmonic coefficients c_i that show colour based on the view. The scene is set up using the sparse SfM point cloud, which has one Gaussian for each 3D point that was found. This initialization is very important. As shown in Figure 3, random initialization creates a lot more background artifacts and floaters than SfM-based initialization.

Figure 3 (from Kerbl et al., 2023): Effect of initialisation strategy. SfM-based initialisation (bottom) produces significantly fewer background floaters and better overall quality compared to random initialisation (top).

2.3 Anisotropic Covariance Modelling

A key novelty that lies within 3DGS is the use of a full anisotropic covariance matrix instead of the isotropic (spherical) one. This allows the primitives to model elongated, planar, or needle-like objects which makes the surface representations highly accurate from any angle of observation. To preserve the covariance positive semi-definiteness at all stages of optimization, a separate q and s optimization is employed, and is:

$$\boldsymbol{\Sigma} = \mathbf{R}\boldsymbol{S}\boldsymbol{S}^T\boldsymbol{R}^T$$

where R ∈ SO(3) (3×3 matrix representing pure rotation in 3D space) is a rotation matrix parameterised by a unit quaternion q, and S = diag(s_1, s_2, s_3) is a diagonal scaling matrix. The optimal flat surface Gaussian, such as the painted body panels of the Caterpillar machine, would have two large scale values along the surface axes and one small scale perpendicular to it. In rendering, the 3D covariance is projected to the camera coordinate system using

$$\Sigma' = JW\Sigma W^T J^T$$

where W is the world-to-camera transform and J is the Jacobian of the projection matrix.

Figure 4 from Kerbl et al., 2023 demonstrates the anisotropic Gaussians obtained after training by reducing them to 60%. It can be seen that the use of anisotropy allows compact modeling of complex geometry which is clearly demonstrated.

Figure 4 (from Kerbl et al., 2023): Trained 3D Gaussians visualised at 60% scale (right) compared to the full render (centre). The elongated shapes demonstrate how anisotropic covariance allows compact surface-aligned representation of complex geometry

The quantitative analysis presented in Table 3 shows the importance of anisotropy: substituting the full covariance matrix with a spherical one results in a decrease in PSNR value of around 0.82 dB at iteration 30,000 as well as the loss of fine detail, see Figure 5.

Figure 5 (from Kerbl et al., 2023): Ablation of anisotropic covariance. The full anisotropic model (centre) significantly outperforms the isotropic baseline (right), particularly for thin structures and fine surface detail.

2.4 Adaptive Density Control

The sparse SfM initialization starts the training process, and the Gaussian set grows or shrinks as the optimization goes on. Every 100 iterations, Gaussians with big view-space positional gradients, which show that something is under- or over-reconstructed, are either cloned (for small regions that are under-reconstructed) or split (for big regions that are over-reconstructed). Every 3,000 iterations, low-opacity Gaussians are cut down. This is what the densification scheme looks like in Figure 6.

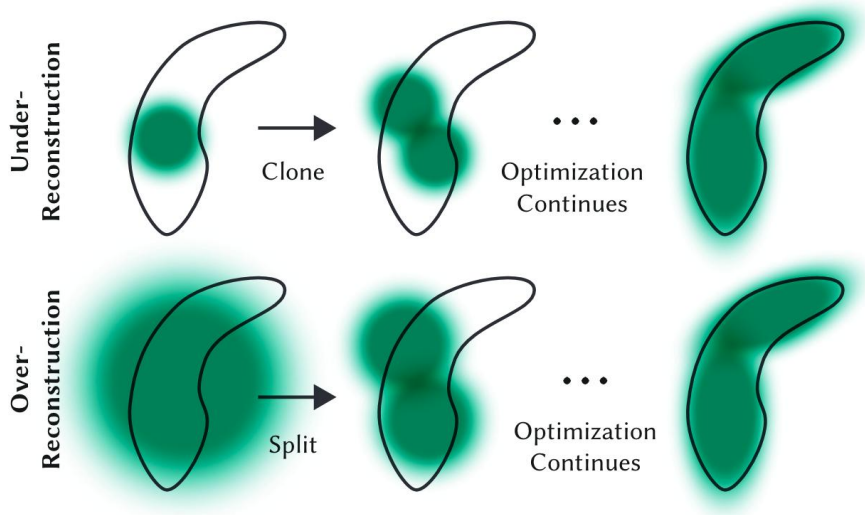

Figure 6 (from Kerbl et al., 2023): Adaptive densification of 3D Gaussians. When under-reconstructed (top part), small Gaussians duplicate themselves and shift along the gradient direction. In case of over-reconstruction (bottom part), large Gaussians divide into two smaller.

2.5 Spherical Harmonics and View-Dependent Colour

Using Spherical Harmonic (SH) basis functions up to degree 3, each Gaussian's colour is shown. In total, there are 16 coefficients per colour channel (totalling 48). This allows smooth colour transition depending on the viewing angle which is important for modelling highlights on reflective materials. Spherical harmonics coefficients are added gradually in order of increasing frequency. At first, only the diffuse coefficient is estimated. Then, every 1,000 iterations one SH band is added until the maximum frequency is reached.

2.6 Differentiable Tile-Based Rasterization

In rendering, the images are divided into 16×16 tile regions. All the primitives are sorted according to their distance from the camera with the help of GPU-based Radix sort algorithm. Rasterization is implemented through alpha blending

$$C = \sum_{i=1}^{N} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_i)$$

As a result, the whole pipeline is differentiable and allows propagating gradients through the rasterizer to all Gaussian parameters. In total, the cost function consists of L1 and D-SSIM parts:

$$L = (1 - \lambda) L_1 + \lambda L_{D-SSIM} \quad ; \lambda = 0.2.$$

Unlike previous works, there is no upper limit on the number of primitives that may receive gradients. Therefore, the scene may contain as many Gaussians as necessary to accurately represent its depth complexity.

3. Dataset and Preprocessing

The dataset consists of 150 pictures of a Caterpillar 950F wheel loader taken outdoors in the sun without control. All the pictures were taken from different angles, heights, and distances. This was a real test in the field because there was no control on lights, background removal, or masks usage.

Property	Value
Subject	Caterpillar 950F Wheel Loader
Number of Images	150
Environment	Outdoor, natural lighting, no background mask
Image Format	JPG
COLMAP Camera Model	OPENCV (radial + tangential distortion)
Dataset Hosting	Kaggle (suhanimanishacharya/caterpillar-images)

Table 1: Dataset summary

There are some problems with rebuilding the machine after rendering. The large yellow-painted body panels have the same texture, which makes it harder for SIFT to differentiate. The rubber tires, bucket linkage, and cab frame all add a lot of texture. The glass cab windows are clear and very shiny, and the hydraulic parts have strong metallic highlights that depend on the angle of view. The unmasked sky background brings up the known unbounded-scene floater problem (Kerbl et al.).

4. Pipeline Implementation

4.1 COLMAP Stages

The convert.py script from the official repository ran the preprocessing pipeline, which included four steps in order:
(1) SIFT feature extraction using the OPENCV camera model
(2) exhaustive feature matching across all 11,175 image pairs
(3) incremental Structure-from-Motion with global bundle adjustment
(4) image un-distortion converting from OPENCV to a pinhole camera model.
Because of a driver compatibility problem in the cloud environment, GPU-accelerated feature extraction was not possible. Instead, CPU extraction was used.

SfM Metric	Value	Interpretation
Images Registered	150 / 150	100% perfect coverage
Sparse 3D Points	93,611	Good density for initialisation
Bundle Adjustment Iterations	5	Fast convergence
Final Reprojection Error	0.454 px	Excellent: sub-pixel accuracy
SfM Wall Time	~9.1 minutes	

Table 2: COLMAP Structure-from-Motion results

4.2 3DGS Training

Training took place on the Kaggle cloud with two NVIDIA Tesla T4 GPUs (15 GB VRAM each) and the default 30,000-iteration schedule. To keep VRAM from running out, image data was moved to the CPU (- data_device cpu). Adaptive density control was always on: Gaussians were densified every 100 iterations and cut down every 3,000 iterations. The model was evaluated at iterations 7,000 and 30,000.

Component	Specification
GPU	2x NVIDIA Tesla T4 (15 GB VRAM each)
Platform	Kaggle Free Tier
Training Iterations	30,000
Total Training Time	~1 hour 27 minutes
Framework	PyTorch 2.10.0 + CUDA 12.8
Data Device	CPU (data device CPU)

Table 3: Training environment

5. Results and Analysis

The trained model was evaluated using PSNR and L1 photometric loss at two checkpoints. PSNR = $10 \log_{10}(MAX^2 / MSE)$, where higher values indicate better reconstruction fidelity.

Iteration	L1 Loss	PSNR (dB)	Quality Band	Time
7,000	0.02768	27.85 dB	Good	~16 min
30,000	0.01746	31.53 dB	Excellent	~87 min
Change	−0.01022	+3.68 dB	36.9% loss reduction	—

Table 4: Training results at evaluation checkpoints

For an outdoor industrial scene, a final PSNR of 31.53 dB is a good score. The 3.68 dB increase from 7K to 30K indicates that fine structures keep getting better in the later stages of training, particularly the tyre tread grooves and hydraulic arm surfaces. In Figure 7 below, the checkpoints at 7K and 30K are compared.

Figure 7 Figure 7: Comparison of rendered output at iteration 7,000 (left, PSNR = 27.85 dB) and iteration 30,000 (right, PSNR = 31.53 dB). Extended training visibly reduces background artefacts and improves surface detail on the machine body and structural components

Figure 8: two (out of 150) input images to represent the input data set (images)
To view full dataset input: https://drive.google.com/drive/folders/1q3S9fHy_vg_yWHUyLhKT_CvNqdf3K3mm?usp=sharing

Figures 9 to 11 show the rendered 3D model of the Caterpillar 950F from different angles, which were generated from the point cloud using the SuperSplat viewer.ply (at iteration 30,000).

Figure 9: Three quarter front renders of the Caterpillar 950F reconstruction at 30k iterations (PSNR = 31.53 dB). The entire structure is visible, proving a successful camera alignment and dense Gaussian cover.

Figure 10: Caterpillar 950F reconstruction at side-profile angle. Tire treads are perfectly replicated due to the abundance of SIFT features. The appearance of needle-like artefacts near metal hydraulic connections shows dependence on viewing direction.

Figure 11: Operator cab close-up. The 'CATERPILLAR' brand text is still readable, which shows a high level of resolution achieved. Glass parts are partially transparent and blurred, illustrating a failure of the Lambertian surface assumption of traditional 3DGS.

The comparison of Figures 9-11 suggests a specific hierarchy of spatial quality: opaque textured surfaces (tires, body panels) show the best quality; thin structures (hydraulic arms, mirrors) show moderate quality but have some floaters; and finally, transparent and specular surfaces (glass, polished metal) show the most obvious artefacts.

6. Surface-Specific Observations

The Caterpillar 950F presents a diverse range of surface materials within a single subject, providing a good test input for evaluating 3DGS behaviour on industrial scenes.

6.1 Painted Metal Body Panels

Despite the generally smooth texture of the body panels painted yellow, they were reconstructed fairly well. The anisotropic covariance mechanism managed to account for the lack of high-frequency detail due to the presence of large, flat Gaussians fitting the curves of the body panel very well. The spherical harmonics-based colour model succeeded in capturing small shading effects due to ambient occlusions and direct lighting. The embossed labels showing "CATERPILLAR" and "950F" were visible, implying sub-centimetre spatial resolution.

6.2 Rubber Tyres

The tire tread pattern was one of the components with the best reconstruction quality. The high-frequency texture gave many good matches with the SIFT algorithm, allowing for placing dense Gaussians by adaptive cloning ("under-reconstruction" path in Figure 6). The dark, diffuse surface made from rubber had little view-dependent changes, making it easier to optimize spherical harmonics.

6.3 Glass Cab Windows

The most difficult element to reconstruct is the glass windows. Firstly, it is because glass defies two important assumptions: (i) glass is translucent while the formula of alpha compositing allows Gaussians either to be fully or partly opaque and (ii) glass reflects and refracts light differently depending on the view direction, and therefore its appearance strongly depends on the position of the viewer. The resulting reconstruction has a blurred semi-transparent zone consisting of several Gaussians inside the cab. This is a known failure case for surfaces whose appearance changes dramatically with viewing angle - the reconstruction cannot assign a single, consistent colour to Gaussians in that region, resulting in unstable, view-dependent flickering artefacts when the camera moves around the glass.

6.4 Hydraulic Arms and Metallic Parts

The geometry of the cylinders was recovered successfully. However, their specularity interfered with colour optimization. It was impossible to consistently assign the bright and viewpoint-dependent speckles to particular Gaussians, and therefore they appeared in the form of needle-like floating Gaussians emerging from metal joints. This problem relates to the "elongated artefact" problem described in the limitations of the Kerb algorithm. The problem occurs when Gaussians cannot define the stable colour of objects with viewpoint-dependent appearance.

6.5 Sky and Background

The most problematic area regarding floaters was the sky region. It had arbitrary-depth Gaussians in the background which did not have any constraints other than distance limits. Without background modelling or scene contraction, Gaussians in the sky cannot be effectively pruned since they might still be useful for some views. Pruning can be done easily if the background is pre-masked before training.

6.6 Top and Elevated Views

The top-down view of the machine shows a lot of geometric distortion, especially around the cab roof, the upper hydraulic arms, and the boom structure. This is directly related to the training dataset: only 150 of the 330 images that were available were used (RAM and processing limitations), and most of these images were taken from eye-level or low angles. The machine's top surfaces were only visible in a few training views, which didn't give the Gaussian covariance enough multi-view constraints to find the right orientation. Instead of making flat anisotropic Gaussians that lined up with the roof surface, the optimizer made Gaussians with random orientations that weren't well-constrained. These can be seen as the swirling distortion artifacts in the top-angle renders (Shown in Figure 11).

Surface Type	Quality	Key Challenge
Painted body panels	Excellent	Low texture - compensated by anisotropy
Rubber tyres	Excellent	None - rich texture, diffuse material
Glass cab windows	Poor	Transparency + specular reflection
Hydraulic metal arms	Moderate	View-dependent specular highlights
Ground terrain	Good	Stable diffuse, well-textured
Sky/background	Poor	Unbounded, no consistent depth
Top View	Poor	Top surfaces of the machine were visible in very few training views

Table 5: Per-surface reconstruction quality

7. Limitations

- Surfaces that are clear and reflective are not modelled well. There is no way for the alpha-compositing renderer to do refraction or transmission. It is noted that the method has a current limitation: glass and polished metal need extensions beyond the base 3DGS formulation.
- The sky background that had no limits made floater artifacts. In regions without scene contraction or background models, Gaussians are positioned at random depths. This is a known problem for outside-in captures with gaps in angular coverage.
- PSNR was only measured on training views. Since there was no held-out test split, the reported 31.53 dB may be too high for generalization quality. Every eighth photo is used as a test set, not for evaluation.
- Image coverage was only available from outside views that could be reached. The undercarriage and rear hydraulics were seen less often, which made the Gaussian density lower in those areas.
- Exhaustive COLMAP matching scales as $O(N^2)$ when the number of images increases. It took 130 minutes for 150 images, but for 500 or more images, vocabulary-tree-based matching would be needed.
- The trained model is too big (488 MB) to be used on devices with limited memory. The memory use is significantly higher than NeRF-based solutions, but compression is a the best possible way.
- Only 150 of the 330 images that were available were used for training, and there wasn't much coverage from above or below. People could see the Caterpillar's roof, upper hydraulic boom, and top cab structure from far fewer angles than from the sides and front. This directly caused the visible distortion in top-angle rendered views because the Gaussian optimizer didn't have enough multi-view constraints to accurately resolve geometry in these areas that weren't well-observed. If you use all 330 pictures with planned top-angle shots, this distortion will decrease significantly.

8. Future Improvements

- Masking using the Segment Anything Model (SAM) would enable better separation of background and segmentation of the subject before the training step. It would get rid of sky floaters and make sure that Gaussian budget goes towards reconstructing the subject.
- Use normal-based covariance initialization by computing normals of the SfM point cloud prior to training, and orienting the smallest scale axis of each Gaussian orthogonal to the surface normal. Doing this would help speed up convergence for planar and cylindrical surfaces, possibly leading to higher PSNR earlier on iteration 7,000.
- Add more images to the total amount by raising it up to 300-500 to achieve a 360-degree coverage of angles, including the undercarriage and high-angle views.
- Implement the held-out test set, which takes every eighth image, will yield a reliable and unbiased PSNR measurement the model has never seen during the training, against which we will be able to compare our results.
- Explore 2D Gaussian Splatting method (Chen et al., 2024). This technique uses oriented discs to achieve better reconstruction performance and cleaner meshes, making it suitable for integration into CAD workflow.
- Use the Spherical Harmonics representation of the colours in terms of the coordinates of the local frame of reference of each individual Gaussian, rather than globally. The SH representation would help achieve more stable colours for metallic objects such as the hydraulic joint.
- Apply the SuGaR (Guédon and Lepetit, 2024) mesh extraction technique to translate the Gaussian-based representation into a mesh form usable in conventional engineering programs.
- In addition to pruning the Gaussians based on occupancy, add a depth-based criterion that will prune the Gaussians whose computed normals do not match the normals of the other Gaussians in the neighbourhood. Depth-based pruning can help eliminate the floating needles observed in the joint surfaces.
- Implement monocular depth priors as additional supervision signals in areas without texture or well-defined surfaces, where photometric loss alone is not adequate for geometrical constraints.

9. Conclusion

The experiment conducted here successfully implemented the full 3D Gaussian Splatting pipeline, from image acquisition via COLMAP preprocessing to Gaussian optimization, producing a highly accurate reconstruction of the Caterpillar 950F wheel loader. The pipeline successfully registered 150 images with 100% accuracy, achieved a reprojection error of 0.454 px, and generated a PSNR of 31.53 dB after 30,000 iterations. The anisotropic covariance scheme showed promising results for the large planar surfaces of the machine. The spherical harmonic colour model performed well in capturing the appearance of painted and metallic surfaces at different views. Limitations encountered, like artifacts arising from glass transparency, sky areas, and reflections on metallic surfaces (e.g., needle artifacts), are consistent with those presented in Kerbl et al., 2023 paper, thus demonstrating that the implementation faithfully replicates the behaviour of the algorithm. 3D Gaussian Splatting is a mature technology with great potential for reconstructing objects and can create photorealistic digital twins from unordered image sets in approximately two hours on affordable cloud GPU resources. These findings confirm its industrial applicability while simultaneously showing the need for modifications to improve performance on complex machinery, such as background masking and transparent surface treatment.
GitHub Repository: https://github.com/suhaniacharya22/3D_caterpillar-machine_Suhani-Acharya

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